


USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

First/Second Semester B.E. (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CY110/20CY110 - ENGINEERING CHEMISTRY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Derive the Nernst equation for single electrode potential.	6	L*2	1	1
b)	Explain the construction and working of Calomel electrode.	7	L2	1	1
c)	What are ion selective electrodes? Explain the determination of pH of a solution using glass electrode.	7	L2	1	1
2. a)	Explain electrochemical theory of corrosion taking iron as example.	7	L2	1	1
b)	What is differential aeration corrosion? Explain water line and pitting corrosion.	7	L2	1	1
c)	Explain the following. (i) Anodizing (ii) Galvanization.	6	L2	1	1
3. a)	Explain the construction and working of Nickel-Metal hydride battery.	7	L2	1	1
b)	Explain the electroless plating of copper on PCB.	7	L2	1	1
c)	Explain the construction and working of Methanol-Oxygen fuel cell.	6	L2	1	1
Unit – II					
4. a)	Explain the free radical mechanism of addition polymerisation using suitable example.	7	L2	2	1
b)	Give the synthesis and application of (i) Polymethyl methacrylate (PMMA) (ii) Polycarbonate	6	L2	2	1
c)	What are adhesives? Explain the synthesis and application of epoxy resin.	7	L2	2	1
5. a)	Explain the determination of hardness of water sample using EDTA.	7	L2	2	1
b)	What is sewage? Explain sewage treatment methods.	7	L2	2	1
c)	Explain the desalination of water by electro-dialysis method.	6	L2	2	1
6. a)	Explain the synthesis of nanomaterial by Sol-gel method.	7	L2	2	1
b)	What are elastomers? Explain the synthesis and application of butyl rubber and silicone rubber.	7	L2	2	1
c)	Explain softening of water by ion exchange method.	6	L2	2	1
Unit – III					
7. a)	Explain bomb calorimetric method of determining calorific value of a solid fuel.	7	L2	3	1
b)	What is reformation? Explain any three-reformation process.	7	L2	3	1
c)	Explain Nematic and Chiral nematic mesophases of liquid crystal.	6	L2	3	1
8. a)	What is cracking? Explain the process of fluidized bed catalytic cracking of petroleum.	7	L2	3	1
b)	On burning 0.96 grams of a solid fuel in bomb calorimeter, the temperature of 3500 grams of water increased by 2.7°C. Water equivalent of calorimeter and latent heat of steam are 385 grams and 587 Calories/grams respectively. If the fuel contains 5% H ₂ . Calculate its gross and net calorific values.	7	L3	3	2
c)	Explain the electro optic effect of liquid crystals.	6	L2	3	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
